ROSMASTER-M3Pro-MCP Feature Summary

This document summarizes all MCP tool features registered in `mcp_action.py`, helping users quickly understand the function, parameters, and usage of each tool.

Feature Overview

No.	Tool Name	Function Description
1	MoveWithSpeed	Control the robot's omnidirectional chassis speed
2	Move	Control the robot chassis to move a specified distance
3	GetCurrentArmJointAngles	Retrieve the robotic arm's current joint angles
4	GetEndEffectorPose	Retrieve the end-effector's pose
5	ControlSixArmJoint	Simultaneously control the arm's 6 joints
6	ControlSingleArmJoint	Control a single arm joint to a specified angle
7	Place	Place an object at specified coordinates
8	Pick	Grasp an object
9	AdjustCameraView	Adjust the camera's viewpoint
10	SeeWhat	Capture a camera image and return the save path
11	InitArmPose	Restore the arm to its preset initial pose
12	GetBbox	Detect the bounding box coordinates of a target object
13	GetPlacePoint	Retrieve placement point coordinates and check reachability
14	AdjustChassisFitArmRange	Quickly adjust the chassis so the target point is within the arm's workspace
15	RecordMapLocation	Save the current location to a map mapping file
16	GetMapMapping	Retrieve the mapping relationship between location names and symbols
17	Navigation	Navigate to a specified location using a target symbol
18	GetWasteRecognitionResults	Retrieve waste recognition results from the YOLO model
19	GraspWaste	Grasp a waste object
20	PlaceWaste	Place a waste object
21	TTS	Robot speech synthesis
22	Rotate	Rotate in place by a specified angle
23	TargetTrack	Visually track a target object
24	GetTargetDist	Calculate the distance between the target and the robot chassis
25	GraspArUcoTarget	Grasp an AprilTag target with a specified ID

No.	Tool Name	Function Description
26	GetAprilTagIDs	Retrieve AprilTag IDs within the field of view
27	FollowLine	Follow a line of a specified color

Detailed Function Descriptions

1. MoveWithSpeed

Function: Controls the robot's omnidirectional chassis movement using velocity commands.

Parameters:

- `linear_x`: Linear velocity in the X direction (Range: -2.5 ~ 2.5; Default: 1.5)
- `linear_y`: Linear velocity in the Y direction (Range: -2.5 ~ 2.5; Default: 1.5)
- `angular_z`: Angular velocity around the Z axis (Range: -2.5 ~ 2.5; Default: 1.5)
- `duration`: Duration of movement (in seconds; Default: 1.0)

Return Value: `common_response`

---

2. Move

Function: Controls the robot's chassis to move a specified distance.

Parameters:

- `distance`: Movement distance (in meters)
- `direction`: Movement direction
- `'forward'`: Move forward
- `'backward'`: Move backward
- `'left'`: Move laterally to the left
- `'right'`: Move laterally to the right

Return Value: `common_response`

---

3. GetCurrentArmJointAngles

Function: Retrieves the current joint angles of the robotic arm.

Parameters: None

Return Value: `ArmJointAngles`

- `arm_joint_1` ~ `arm_joint_5`: Angles for robotic arm joints 1 through 5.
- `gripper`: Gripper angle.

---

4. GetEndEffectorPose

Function: Retrieves the pose (position and orientation) of the end effector (comprising the gripper and camera).

Parameters: None

Return Value: `EndEffectorPose`

- `x`, `y`, `z`: Position coordinates.
- `roll`, `pitch`, `yaw`: Orientation angles.

---

5. ControlSixArmJoint

Function: Simultaneously controls all 6 joints of the robotic arm (the 6th joint controls the gripper angle).

Parameters:

- `arm_joint_1`: Joint 1 angle (Range: 0 ~ 180).
- `arm_joint_2`: Joint 2 angle (Range: -5 ~ 180).
- `arm_joint_3`: Joint 3 angle (Range: -10 ~ 180).
- `arm_joint_4`: Joint 4 angle (Range: -50 ~ 180).
- `arm_joint_5`: Joint 5 angle (Range: 0 ~ 180).
- `arm_joint_6`: Gripper angle (Range: 0 ~ 180; 180 = fully open, 0 = fully closed).
- `runtime`: Duration of the movement (in milliseconds; Default: 1500; Range: 0 ~ 2000).

Return Value: `common_response`

---

6. ControlSingleArmJoint

Function: Sets a single robotic arm joint to a specified angle.

Parameters:

- `arm_joint_id`: Joint ID (1-6; corresponds to robotic arm joints 1-6, where Joint 6 is the gripper).
- `angle`: Target angle (For Joint 6... - `gripper_range`: Gripper range (0 ~ 180)
- `runtime`: Motion duration (milliseconds; default: 1500, range: 0 ~ 2000)

Return Value: `common_response`

---

7. Place

Function: Places an object at the specified coordinates

Parameters:

- `place_x`: X-coordinate (meters)
- `place_y`: Y-coordinate (meters)
- `place_z`: Z-coordinate (meters)

Return Value: `common_response`

8. Pick

Function: Grabs an object.

Parameters:

- `x1`, `y1`: Coordinates of the top-left corner of the target object.

- `x2`, `y2`: Coordinates of the bottom-right corner of the target object.

Return Value: `common_response`

---

9. AdjustCameraView

Function: Adjusts the camera view (pitch and yaw angles).

Parameters:

- `pitch`: Pitch angle adjustment (degrees; positive = up, negative = down; recommended step size: 5–10 degrees; default: 0).
- `yaw`: Yaw angle adjustment (degrees; positive = right, negative = left; recommended step size: 5–10 degrees; default: 0).

Return Value: `dict` or `common_response`

- Returns `{"image": adjusted_image}` upon success.
- Returns `common_response` upon failure.

---

10. SeeWhat

Function: Captures an image from the camera and returns the path where the image is saved.

Parameters: None.

Return Value: `dict` or `common_response`

- Returns `{"image": image_save_path}` upon success.
- Returns `common_response` upon failure.

---

11. InitArmPose

Function: Restores the robotic arm to its preset initial pose.

Parameters: None.

Return Value: `common_response`

---

12. GetBbox

Function: Detects and returns the bounding box coordinates of a target object based on a description.

Parameters:

- `query`: A description of the target object at its current location, including its visual features.

Return Value: `BoxLocalization` or `common_response`

- Returns `BoxLocalization(bbox=bounding_box, verify_image=verification_image_path)` upon success.
- Returns `common_response` upon failure.

---

13. GetPlacePoint

Function: Retrieves the coordinates for a placement point and checks whether it lies within the robotic arm's reachable workspace.

Parameters:

- `query`: Query description.

Return Value: `PlacePoint` or `common_response`

- Returns upon success: `PlacePoint(info=info, target_point=target_point, place_verify_image=place_verify_image_path)`
- Returns `common_response` upon failure.

---

14. AdjustChassisFitArmRange

Function: Quickly adjust the mobile chassis so that the target point falls within the robotic arm's workspace.

Parameters:

- `x`, `y`, `z`: Target coordinates (meters)

Return Value: `common_response`

---

15. RecordMapLocation

Function: Save the current location to the map mapping file.

Parameters:

- `name`: Location name
- `symbol`: Location symbol

Return Value: `common_response`

---

16. GetMapMapping

Function: Retrieve the mapping relationship between all location names and their corresponding symbols.

Parameters: None

Return Value: `common_response` or `str`

- Returns the mapping relationship string upon success.
- Returns `common_response` upon failure.

---

17. Navigation

Function: Navigate to a specified location using its target point symbol.

Parameters:

- `location`: The symbol corresponding to the location in the map mapping file.

Return Value: `common_response`

---

18. GetWasteRecognitionResults

Function: Retrieve waste recognition results from the YOLO model.

Parameters: None

Return Value: `waste_recognition_result` or `common_response`

- Returns the waste recognition results upon success.
- Returns `common_response` upon failure.

---

19. GraspWaste

Function: Grasp a waste object.

Parameters:

- `waste_name`: Name of the waste object (obtained from the waste recognition results).

Return Value: `common_response`

---

20. PlaceWaste

Function: Place a waste object.

Parameters: None

Return Value: `common_response`

---

21. TTS

Function: Robot Text-to-Speech (TTS) synthesis.

Parameters:

- `text`: The text content to be synthesized.

Return Value: `common_response`

---

22. Rotate

Function: Rotate in place by a specified angle.

Parameters:

- `angle`: Rotation angle (degrees; negative = clockwise/right, positive = counter-clockwise/left).

Return Value: `common_response`

---

23. TargetTrack

Function: Visually track a target object

Parameters:

- `x1_or_cmd`:
- String `"cancel"`: Stop current tracking
- Integer: Serves as the x1 coordinate of the target tracking bounding box
- `y1`: y-coordinate of the target object's top-left corner (Optional)
- `x2`: x-coordinate of the target object's bottom-right corner (Optional)
- `y2`: y-coordinate of the target object's bottom-right corner (Optional)

Return Value: `common_response`

---

24. GetTargetDist

Function: Calculate the distance between the target and the robot chassis based on an object description

Parameters:

- `query`: A description of the target object's position and visual features within the current field of view (String)

Return Value: `common_response`

---

25. GraspArUcoTarget

Function: Grasp an AprilTag target with a specified ID

Parameters:

- `target_id`: The ID of the AprilTag to be grasped (Integer)

Return Value: `common_response`

---

26. GetAprilTagIDs

Function: Retrieve the IDs of AprilTags currently within the field of view

Parameters: None

Return Value: `common_response`

---

27. FollowLine

Function: Drive along a colored line on the ground

Parameters:

- `color`: The color of the line to follow; possible values: `"red"`, `"green"`, `"blue"`, `"yellow"` (String)

Return Value: `common_response`

Function Categories

Chassis Control

- MoveWithSpeed
- Move
- Rotate
- AdjustChassisFitArmRange
- Navigation

Robotic Arm Control

- GetCurrentArmJointAngles
- GetEndEffectorPose
- ControlSixArmJoint
- ControlSingleArmJoint
- InitArmPose

Grasping Operations

- Pick
- Place
- GraspWaste
- PlaceWaste

Vision Functions

- AdjustCameraView
- SeeWhat
- GetBbox
- GetPlacePoint
- TargetTrack
- GetWasteRecognitionResults
- GetTargetDist
- GetAprilTagIDs
- GraspArUcoTarget
- FollowLine

Mapping & Navigation

- RecordMapLocation
- GetMapMapping
- Navigation

Speech Synthesis

- TTS